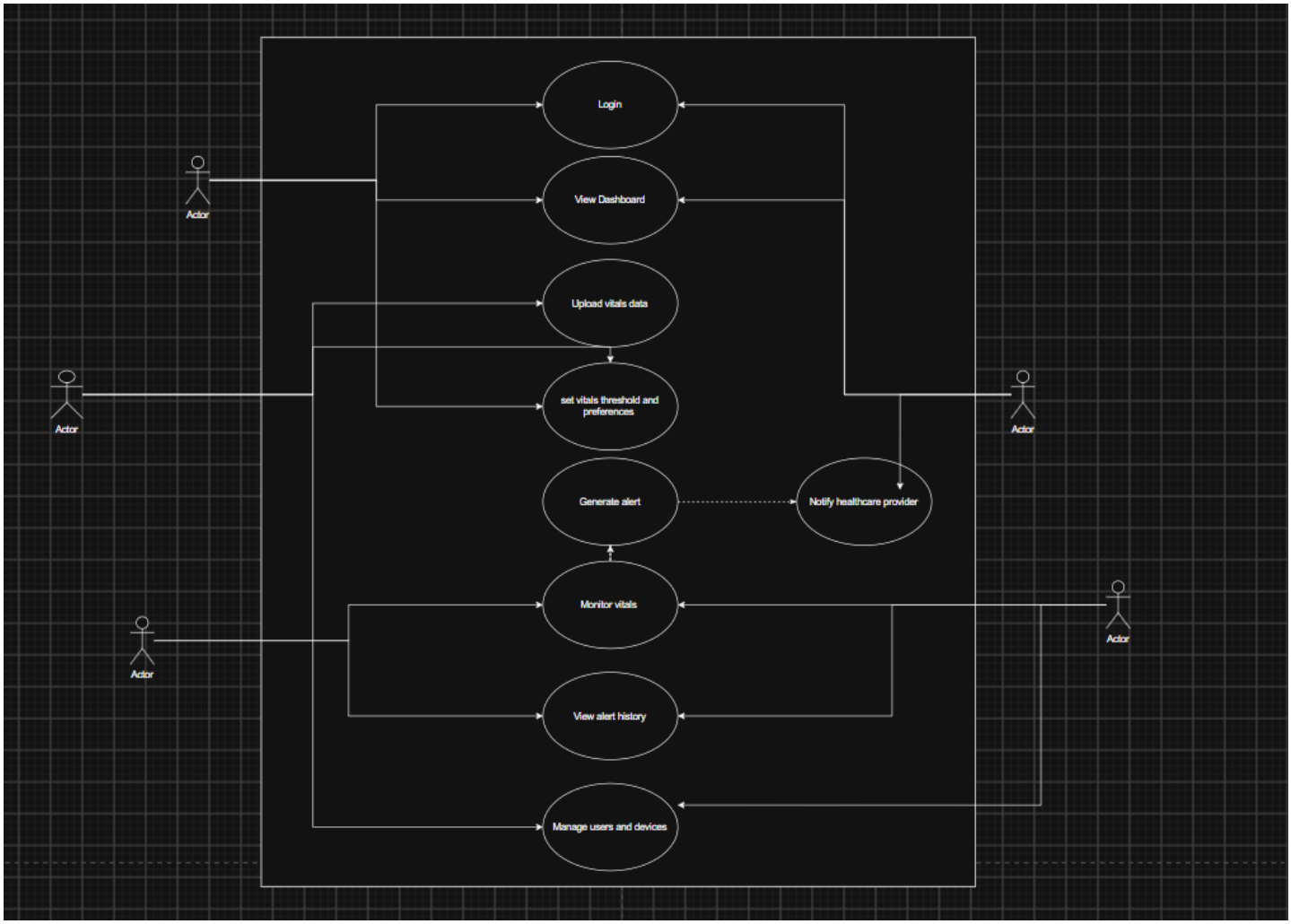
Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow a patient to pair a wearable device (or enter vitals manually) and receive continuous or on-demand vital-sign readings (HR, SpO_2, BP, temperature, RR).	High	After successful pairing, a new reading appears on the patient dashboard within 10 seconds of device transmission or manual save; all five vital types are accepted and stored.	Core data-acquisition capability required for any monitoring or alerting.
FR-002	Functional	The system shall automatically evaluate each incoming vital reading against clinician-configured upper and lower thresholds and generate an alert when a threshold is breached.	High	When a simulated reading exceeds a configured threshold, an alert record is created with timestamp, vital type, measured value and threshold within 5 seconds; alert status = “Active”.	Primary clinical value proposition – early detection of deterioration.
FR-003	Functional	The system shall notify the assigned clinician (push notification + in-app alert list) of any critical alert and allow the clinician to acknowledge or escalate the alert.	High	Clinician receives a push notification within 15 seconds of alert generation; acknowledgement changes alert status to “Acknowledged” and is logged with user ID and timestamp.	Ensures timely clinical response and closed-loop accountability.
FR-004	Functional	The system shall provide both patient and clinician with a historical trend view (graph + tabular) of any selected vital sign for the past 24 h / 7 d / 30 d.	Medium	Selecting a vital and time window returns a correctly rendered line chart and table within 3 seconds; data points match the stored readings for that period.	Supports clinical decision-making and patient self-management.
FR-005	Functional	The system shall enforce role-based access control so that a patient can view only their own data and a clinician can view only data of patients assigned to them.	High	A patient login cannot retrieve another patient’s record (HTTP 403); a clinician sees only assigned patients in the patient list and dashboard.	Mandatory privacy and security control for health data.
NFR-001	Non-functional	The system shall deliver any critical alert notification to the assigned clinician within 15 seconds of threshold breach under normal network conditions.	High	In a controlled test with 50 concurrent patients generating one critical alert each, 95 % of notifications arrive at the clinician app within 15 s (measured by server and client timestamps).	Clinical safety requires near-real-time notification of deterioration.
NFR-002	Non-functional	The system shall encrypt all patient vital data in transit (TLS 1.3) and at rest (AES-256) and restrict access according to the principle of least privilege.	High	Penetration test or automated scan confirms TLS 1.3 only, AES-256 at rest, and no privilege-escalation path that allows a patient to read another patient’s data.	Required for regulatory compliance (HIPAA / GDPR) and patient trust.


Login
View Dashboard
Upload vitals data
set vitals threshold and preferences
Generate alert
Notify healthcare provider
Monitor vitals
View alert history
Manage users and devices
Actor
Actor
Actor
Actor
Actor

Lab 1 – Use-Case Flow Document

Remote Patient Vitals Alert & Monitoring App

Use Case: UC-06 – Generate & Deliver Alert

Primary Actor: System (triggered by incoming vital data)
Secondary Actors: Clinician, Patient (optional mild-alert path)
Goal: Detect a threshold breach and reliably notify the responsible clinician within the required time window.

Preconditions

- Patient has at least one wearable device paired or is allowed to enter vitals manually.
- Clinician has been assigned to the patient and has push notifications enabled.
- Alert thresholds for the relevant vital signs have been configured (or system defaults are active).

Main Success Scenario

1. A new vital-sign reading (e.g., Heart Rate = 148 bpm) is received from the patient's wearable device (or entered manually).

2. The system validates the reading format and associates it with the correct patient record.

3. The system retrieves the active upper and lower thresholds for that vital sign and patient.

4. The system compares the measured value against the thresholds and determines that a breach has occurred (e.g., HR > 140).

5. The system creates an Alert record with status = "Active", timestamp, vital type, measured value, threshold value, and severity (Critical / Warning).

6. The system identifies the clinician(s) currently assigned to the patient.

7. The system sends a push notification (and optional SMS/email) to the assigned clinician(s) containing patient ID, vital type, value and severity.

8. The system also places the alert in the clinician's in-app "Active Alerts" list and updates the patient dashboard status indicator.

9. The clinician opens the alert (or receives the push) and acknowledges it; the system records acknowledgement user, timestamp and changes status to "Acknowledged".

10. Use case ends successfully. Alert remains visible in history for audit.

Alternate Flows

4a. Reading is within normal range

4a1. System stores the reading normally.
4a2. No alert is generated.
4a3. Use case ends (normal monitoring path).

7a. Push notification delivery fails (device offline / network error)

7a1. System logs the delivery failure and retries up to 3 times with exponential back-off (5 s, 15 s, 45 s).
7a2. If all retries fail, system escalates by sending an SMS (if configured) and marks the alert as "Delivery Failed – Escalated".
7a3. System Administrator is notified of persistent delivery problems.
7a4. Use case continues; clinician can still see the alert when next online.

9a. Clinician does not acknowledge within the configured escalation window (e.g., 10 minutes)

9a1. System automatically escalates the alert to a secondary clinician or on-call team.
9a2. Alert status is updated to "Escalated".
9a3. Both original and secondary clinicians receive notifications.
9a4. Use case ends when any clinician acknowledges or the escalation policy is exhausted.

Post-conditions

- An Alert record exists with full audit trail (creation, notification attempts, acknowledgement).
- The assigned clinician has been notified (or escalation has occurred).
- Patient's vital history remains intact and queryable.